

Introduction to Queueing Theory

The code used in these notes can be found on Course Resources.

A few notes about the exponential distribution

The exponential distribution is often used to model the inter-arrival times between events. Let a be the average interval between arrivals, then the exponential distribution is defined as,

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

where the arrival rate λ is related to a by $\lambda = \frac{1}{a}$.

Figure 1 shows plots of different arrival rates as expected larger values of λ make smaller values of x (inter-arrival times) more probable, thus on average a smaller value of a .

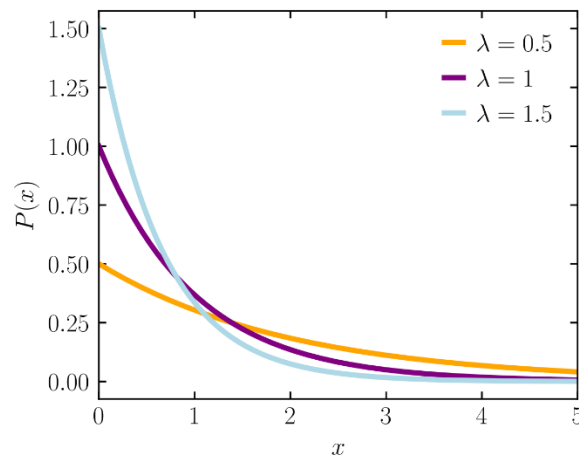


Figure 1: Exponential distribution

For example, if we expected 10 emails in an hour then we have an average interval between events of $a = 6$, i.e., 6 minutes. Therefore, we have an arrival rate $\lambda = \frac{1}{6}$. NOTE: This would mean that we are measure in units of minutes.

If we wanted to use hours, then $a = \frac{1}{10}$, that is emails arrive every tenth of an hour. Therefore, in hours $\lambda = 10$.

Generating inter-arrival times from the distribution (sampling)

To generate a sample from the distribution, we can use the cumulative distribution (see figure 2),

$$F(x) = \begin{cases} 1 - \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

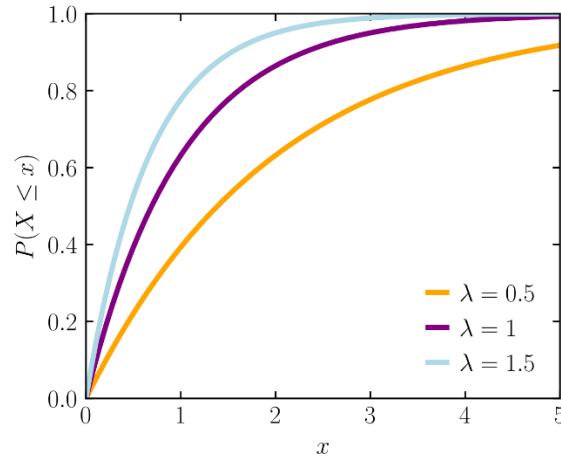


Figure 2: Cumulative exponential distribution

The main idea is to generate a number between 0 and 1 (y) and then read it off the cumulative distribution to generate a value of x .

$$y = F(x)$$

$$F^{-1}(y) = x$$

In the case of the exponential distribution,

$$y = 1 - e^{-\lambda x}$$

$$\ln(1 - y) = -\lambda x$$

$$x = -\frac{1}{\lambda} \ln(1 - y)$$

NOTE: As we are generating random values of $y \in [0,1]$, then $1 - y \in [0,1]$ and also random. Hence we can just use,

$$x = -\frac{1}{\lambda} \ln(y)$$

To confirm this empirically, we can generate 10000 values of x for some average arrival time a and therefore $\lambda = \frac{1}{a}$. Our expectation is that the average of these 10000 values is approximately a .

The following MATLAB code will generate 10000 samples with an average of approximately a .

```
no_samples = 10000
a = 6;
lam = 1/a;

% sample from exponential distribution
Y = rand(1,no_samples);
X = -(1/lam)*log(Y);

% plot the cumulative mean
plot(cumsum(X)./(1:no_samples))
xlabel('No Samples')
ylabel('Cumulative Mean')
```

Figure 3 shows that as we draw more samples, the mean of these samples tends to the average arrival time $a = 6$. Figure 4 plots the 10000 samples as a histogram alongside the exponential distribution with $\lambda = \frac{1}{6}$, i.e., $f(x) = \frac{1}{6}e^{-\frac{1}{6}x}$

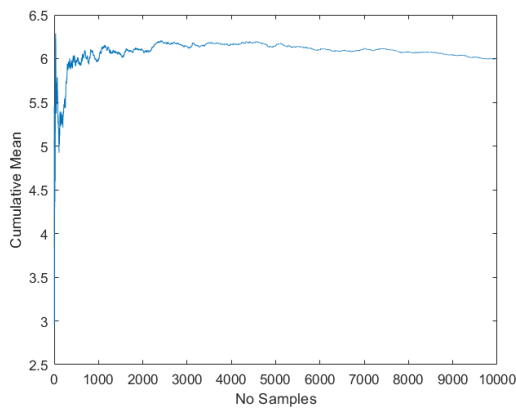


Figure 3: Convergence to the mean

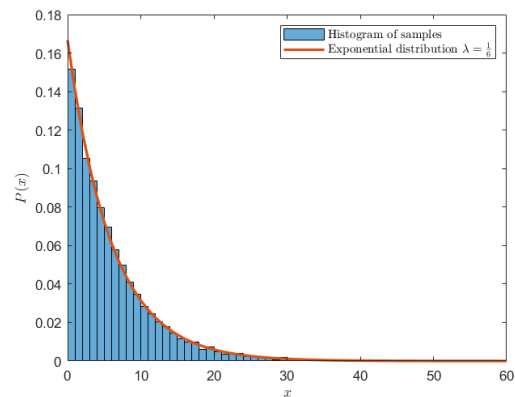


Figure 4: Histogram of samples drawn

M/M/1 Queues

The M/M/1 queue has exponentially distributed times for both the arrival and the service and one server.

Let $f(t)$ denote the inter-arrival times for the arrival of customers (jobs) and a be the average interval between arrivals, then the exponential distribution is defined as,

$$f(t) = \lambda e^{-\lambda t},$$

where $\lambda = \frac{1}{a}$ is the arrival rate.

Similarly let $g(t)$ denote the service times for the serving of customers (jobs) by the server and s be the average service time, then the exponential distribution is defined as,

$$g(t) = \mu e^{-\mu t},$$

where $\mu = \frac{1}{s}$ is the service rate.

Simulation of M/M/1

We fix the value of $s = 1$, i.e., $\mu = 1$, and manipulate the arrival rate λ .

For values of $\lambda < \mu$, we can see from figures 5,6 and 7 that the system settles into a random pattern around its mean.

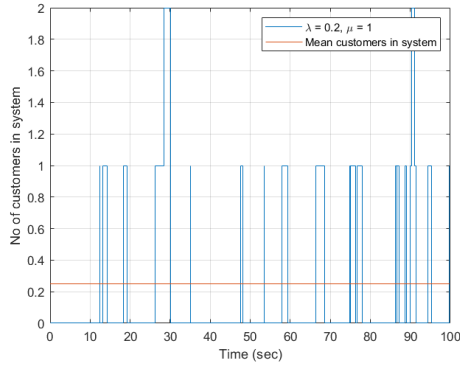


Figure 5: Time 'vs' No. customers in the system

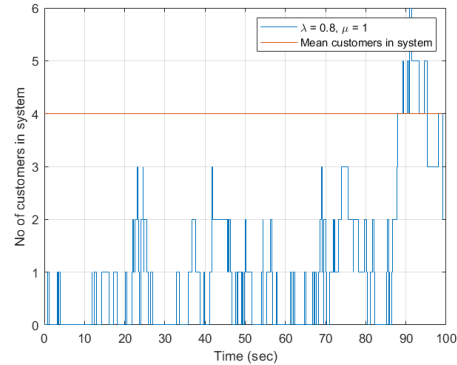


Figure 6: Time 'vs' No. customers in the system

However, as $\lambda \rightarrow 1$ we move towards a situation at which customers will arrive faster than they can be processed, this leads to the queue growing infinitum. Figure 8 plots the situation at the limit, $\lambda = 1$.

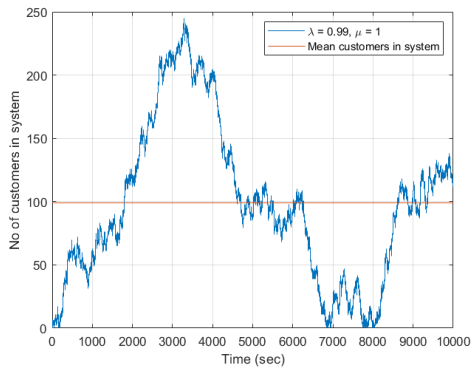


Figure 7: Time 'vs' No. customers in the system

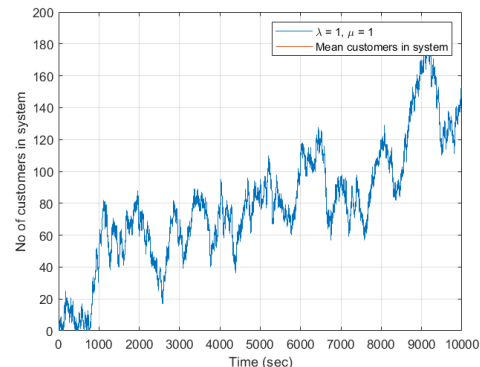


Figure 8: Time 'vs' No. customers in the system

Steady state of a queue

Let $P_t(n)$ denote the probability of having n customers in the system at time t . For $n > 0$ there will be $n - 1$ customers waiting in the queue and 1 customer being served.

We can illustrate what is known Markov chain for the M/M/1 queue by figure 9. Each node represents the number n of customers at a given time t and the edges in are the transition probabilities. In each unit time step, the probability of moving from n to $n + 1$ is λ and thus in a small time step δt the probability is $\lambda \delta t$. Similarly, we have $\mu \delta t$ for n to $n - 1$.

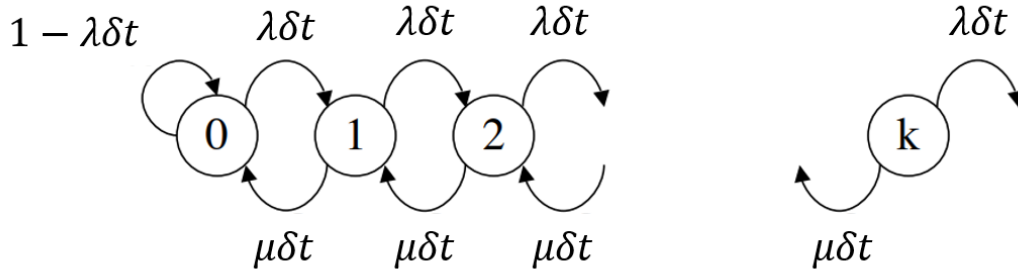


Figure 9: Markov chain for the M/M/1 queue

We now consider the probability of staying in state 0 for a time step δt

$$\begin{aligned}
 P_{t+\delta t}(0) = & P_t(0) \times (\text{chance of nothing happening in the interval}) \\
 & + P_t(1) \times (\text{chance of a service being completed in the interval}) \\
 & + P_t(2) \times (\text{chance of 2 services being completed in the interval}) \\
 & + \dots
 \end{aligned}$$

Reading these off the Markov chain we have,

$$P_{t+\delta t}(0) = P_t(0)(1 - \lambda\delta t) + P_t(1)\mu\delta t + O(\delta t^2) + \dots \quad (1a)$$

$$P_{t+\delta t}(0) = P_t(0)(1 - \lambda\delta t) + P_t(1)\mu\delta t + O(\delta t^2) + \dots \quad (1)$$

Note that $O(\delta t^2)$ denotes a term that is not linear in δt which are representative of multiple events occurring in δt , we will see shortly that these terms will be irrelevant.

We can also compute the probability of staying in the general state n where $n > 1$

$$\begin{aligned}
 P_{t+\delta t}(n) = & P_t(n) \times (\text{chance of nothing happening in the interval}) \\
 & + P_t(n-1) \times (\text{chance of an arrival in the interval}) \\
 & + P_t(n+1) \times (\text{chance a service is completed in the interval}) \\
 & + \text{terms involving multiple events}
 \end{aligned}$$

$$P_{t+\delta t}(n) = P_t(n)(1 - \lambda\delta t)(1 - \mu\delta t) + P_t(n-1)\lambda\delta t + P_t(n+1)\mu\delta t + O(\delta t^2) \quad (2)$$

On rearrangement of (1) and (2) and dividing by δt these become:

$$\begin{aligned}
 \frac{P_{t+\delta t}(0) - P_t(0)}{\delta t} &= \mu P_t(1) - \lambda P_t(0) + O(\delta t) \\
 \frac{P_{t+\delta t}(n) - P_t(n)}{\delta t} &= \lambda P_t(n-1) + \mu P_t(n+1) - (\lambda + \mu)P_t(n) + O(\delta t)
 \end{aligned}$$

As $\delta t \rightarrow 0$,

$$\frac{dP_t(0)}{dt} = \mu P_t(1) - \lambda P_t(0) \quad (3)$$

$$\frac{dP_t(n)}{dt} = \lambda P_t(n-1) + \mu P_t(n+1) - (\lambda + \mu)P_t(n) \quad (4)$$

There is a general solution to this; however, we will concentrate on the steady state of the queue.

That is when $\frac{dP_t(n)}{dt} = 0$, i.e., the probabilities of state changes have reached a stable state. This does not mean that the number of customers in the system is stable, this will still fluctuate. In other words, the long term behaviour of the queue is dominated by the steady-state probabilities.

$$0 = \mu P_t(1) - \lambda P_t(0) \quad (5)$$

$$0 = \lambda P_t(n-1) + \mu P_t(n+1) - (\lambda + \mu)P_t(n) \quad (6)$$

Let $n = 1$ then equation (6) becomes,

$$0 = \lambda P_t(0) + \mu P_t(2) - (\lambda + \mu)P_t(1) \quad (7)$$

Substituting (5) into (7),

$$\mu P_t(2) = \lambda P_t(1)$$

By adding these equations together in sequence $n = 0, 1, 2, \dots, \infty$

$$\mu P_t(n) = \lambda P_t(n-1) \quad (8)$$

Let $\rho = \frac{\lambda}{\mu}$,

$$P_t(n) = \rho P_t(n-1)$$

$$P_t(n) = \rho^n P_t(0) \quad (9)$$

From the axioms of probability

$$\sum_{n=0}^{\infty} P_t(n) = 1$$

From (9),

$$\sum_{n=0}^{\infty} P_t(n) = \sum_{n=0}^{\infty} \rho^n P_t(0) = 1 \quad (10)$$

For $\rho < 1$,

$$\sum_{n=0}^{\infty} \rho^n = \frac{1}{1-\rho}$$

Thus,

$$P_t(0) = \frac{1}{\sum_{n=0}^{\infty} \rho^n} = 1 - \rho \quad (11)$$

Therefore $P_t(n)$ can be computed from (9) and (11),

$$P_t(n) = \rho^n(1 - \rho) \quad (12)$$

Note that this analysis is for $\rho = \frac{\lambda}{\mu} < 1$. If you consider when $\lambda \geq \mu$ then the server cannot complete the arrivals fast enough and thus the queue will grow infinitum.

Expected value of n (mean number of customers in the queue)

Denote the average customers in the system by N , then,

$$N = \sum_{n=0}^{\infty} nP_t(n) = \sum_{n=0}^{\infty} n\rho^n(1 - \rho) = \sum_{n=0}^{\infty} n\rho^n - \sum_{n=0}^{\infty} n\rho^{n+1} \quad (13)$$

To solve this we utilise the following,

$$\frac{d}{d\rho} \sum_{n=0}^{\infty} \rho^n = \sum_{n=0}^{\infty} n\rho^{n-1}$$

And via the chain rule,

$$\frac{d}{dp} \sum_{n=0}^{\infty} \rho^n = \frac{d}{d\rho} \left(\frac{1}{1 - \rho} \right) = \frac{1}{(1 - \rho)^2}$$

Thus,

$$\begin{aligned} \sum_{n=0}^{\infty} n\rho^{n-1} &= \frac{1}{(1 - \rho)^2} \\ \sum_{n=0}^{\infty} n\rho^n &= \rho \sum_{n=0}^{\infty} n\rho^{n-1} = \frac{\rho}{(1 - \rho)^2} \end{aligned}$$

Similarly,

$$\sum_{n=0}^{\infty} n\rho^{n+1} = \frac{\rho^2}{(1 - \rho)^2}$$

From (13)

$$N = \sum_{n=0}^{\infty} n\rho^n - \sum_{n=0}^{\infty} n\rho^{n+1} = \frac{\rho}{(1 - \rho)^2} - \frac{\rho^2}{(1 - \rho)^2} = \frac{\rho}{1 - \rho} \quad (14)$$

$$N = \frac{\rho}{1 - \rho} = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}} = \frac{\lambda}{\mu - \lambda}$$

Average customer time spent in the system

The average amount of time that a customer spends in the system can be obtained from Little's formula

$$N = \lambda T$$

$$T = \frac{N}{\lambda} = \frac{\lambda}{(\mu - \lambda)\lambda} = \frac{1}{\mu - \lambda} \quad (15)$$

Average customer time in queue

The average time spent in the queue is the time spent in the system minus the time spent being served.

The average time being served is defined from the exponential distribution $s = \frac{1}{\mu}$

$$W = T - \frac{1}{\mu} = \frac{1}{\mu - \lambda} - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)} \quad (16)$$

Simulation

The following simulation fixes the average service time $s = 1$ and therefore $\mu = 1$. Arrival rates λ are then generated for $\lambda \in (0,1)$. Therefore, we are inspecting values of $\rho \in (0,1)$.

Figure 10 plots the simulated and analytical results for the mean number of customers in the system as we manipulate ρ . Similarly, figure 11 plots the simulated and analytical results for the mean time a customer spends in the system as we manipulate ρ

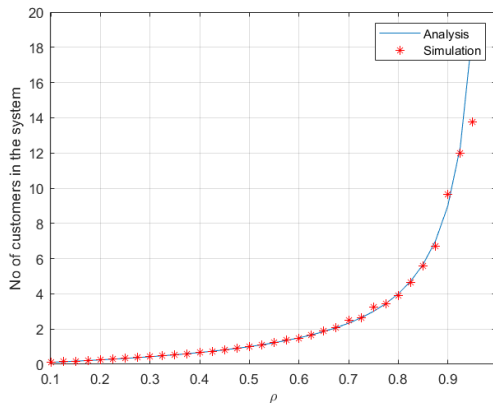


Figure 10: Mean No. customers in the system

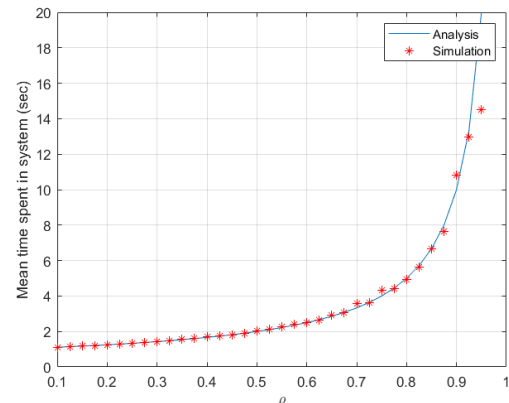


Figure 11: Mean time a customer spends in the system

References

- [1] E. Page, Queueing Theory in OR, Butterworths, 1972.
- [2] J.-B. Seo, "Markov processes and Queueing theory," [Online]. Available: https://web.iitd.ac.in/~jbseo/ell785/Queueing_2017_Fall.pdf.